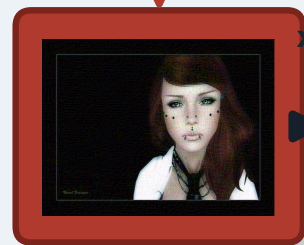


L-SHIELD Defense Pipeline — Latent Space Statistical Clipping

Adversarial Input

$$(\mathbf{x} + \boldsymbol{\delta})$$



$$\mathbf{x}_e = \mathbf{x} + \boldsymbol{\delta}$$

Preprocessing

- 5-bit Quantization
- Gaussian Blur ($\sigma = 1.0$)

$$\tilde{\mathbf{x}}_e$$

VAE
Encoder

$$\mathcal{E}$$

L-SHIELD

Stat. Clip (Ours)

$$\hat{\mathbf{z}}_e = \text{clip}(\mathbf{z}_e, \boldsymbol{\mu} \pm \lambda \boldsymbol{\sigma})$$

Calibration Stats ($\boldsymbol{\mu}, \boldsymbol{\sigma}$)

per-position [offline, pre-computed]

from clean latent representations

Denoising
U-Net

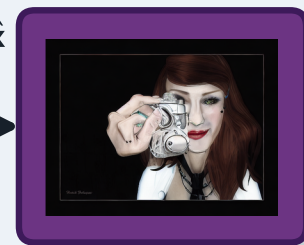
$$\boldsymbol{\theta}$$

VAE
Decoder

$$\mathcal{D}$$

Recovered Output

$$(\hat{\mathbf{x}})$$



Adversarial Input



Preprocess



VAE Enc / Dec



L-SHIELD (Ours)



Denoising U-Net



Clean Output